Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 243 (2024) 1265–1272

Procedia Computer Science

www.elsevier.com/locate/procedia

The 4th International Conference on Machine Learning and Big Data Analytics for IoT Security and Privacy

A Chinese License Plate Recognition System based on OpenCV for complex environments

Huimin Chen[a] ,Xiaofeng Shi[a(✉)], Menghui Liu[a] and Chaozheng Chen[a]

[a]*Department of Computer and Information Engineering, Fuyang Normal University,Fuyang, Anhui, China*

(✉)Corresponding author: shixiaofeng@fynu.edu.cn

Abstract

License plate recognition technology is a key to realizing intelligent traffic management. Nowadays, most of the license plate recognition technologies are only designed for the preferred lighting conditions. But in the natural scenes, the change of the weather and the light intensity throughout the day carry many difficulties to the location and recognition of the license plate system. To solve this problem, we proposed a license plate recognition system based on OpenCV and SVM (Support Vector Machines) in this paper. First, the Sobel Edge Detection algorithm combined with Mathematical Morphology was used to determine the approximate outline of the license plate. And Orthographic Projection was then used to segment the characters in the license plate. Finally, the segmented characters were identified by using SVM. The recognition accuracy of the proposed system is 96.52%. In order to improve the operability of the system, we have designed a Graphical User Interface (GUI) based on PyQt5, which makes the interface more beautiful and simple.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 4th International Conference on Machine Learning and Big Data Analytics for IoT Security and Privacy

Keywords: OpenCV，Support Vector Machines，Sobel Edge Detection，Mathematical Morphology

1.Introduction

Intelligent traffic management is the comprehensive use of data communication, electronic control and artificial intelligence technology in the entire transportation management system, effectively used in transportation, service control and vehicle manufacturing, strengthen the connection between vehicles, roads, users of the three, so as to form a comprehensive transportation system to ensure safety, improve efficiency, improve the environment and save energy. License plate recognition is an important part of traffic intelligence, which mainly includes the following key steps: license plate location, image processing, character segmentation, and character recognition [1].

The key step is to locate the license plate area from the vehicle image. The first step is to correctly detect the license plate area. Researchers have proposed many methods, such as Hough transform and gray threshold. Hough transform is sensitive to image noise, and is easily affected by license plate deformation or noise when detecting the straight line of the boundary of the license plate, and has a large false detection probability. The gray threshold segmentation

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 4th International Conference on Machine Learning and Big Data Analytics for IoT Security and Privacy
10.1016/j.procs.2024.09.149

method is more stable than the linear detection method, but when the image contains areas similar to the gray level of the license plate, the recognition rate will be greatly reduced, and it is no longer applicable.

In view of the above shortcomings, this paper first corrected the original image by perspective transformation before positioning, and proposed a license plate location method based on color features. At the same time, mathematical morphology and sobel algorithm were combined to determine the rough outline, and finally, candidate regions matching the characteristics of the license plate were screened by combining the area and aspect ratio of the external rectangle of the license plate. The new location method avoids the time-consuming problem of algorithms such as Hough transform [2], wavelet transform [3] and single edge detection [4], which is conducive to the rapid location of license plates and the recognition of license plates under complex environmental conditions.

2.The Process Used to Identify License Plate

Fig.1 shows the process used to identify license plate, it mainly includes the following key steps: license plate location, picture processing, character segmentation, character recognition. This paper using Sobel Edge Detection algorithm combined with Mathematical Morphology to determine the approximate outline of the license plate [5], and an improved intensity feature localization algorithm is used to image binarization. It not only improves the accuracy of license plate recognition, but also greatly reduces the data storage capacity, and reduces the follow-up processing time at the same time. It has made a great effort to improve the speed of license plate recognition. In this paper, the upper and lower boundaries of the license plate character region are determined according to the specifications of the license plate, and then the character segmentation method based on the second character is used to segment the license plate character, and finally the SVM algorithm is used to realize the character recognition [6].

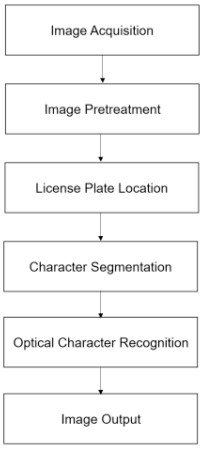


Fig.1. The process used to identify license plate

3.Image Pretreatment

Because the license plate angles taken in the original license plate image are different, the license plate characters cannot be divided for the license plate image with a larger cutting Angle, so it is necessary to correct and transform the license plate image after obtaining the license plate image. Perspective transformation refers to the transformation that makes the perspective surface rotate a certain Angle around the perspective axis according to the perspective rotation law, which takes advantage of the fact that the perspective center, image point and target point are collinear, destroys the original projection beam, and still keeps the projection geometry unchanged on the host surface[9]. The perspective transform can project the image onto A new plane of view, as shown in Fig.2, and the ABC can be transformed into a 'b 'c 'by perspective transformation. We use getPerspectiveTransform function in OpenCV library to perspective transform the license plate area, which can obtain a transformation matrix according to the corresponding positions of the four non-collinear points in the image before and after the transformation. Then the transformed image can be obtained by using the obtained transformation matrix through the warpPerspective function.

To improve the accuracy of license plate location and character recognition in low-quality images, the images are first pre-processed after obtaining the original vehicle image. The aim of this step is to improve the quality of the acquired vehicle image and highlight the key information that needs to be processed and identified in the vehicle image. The commonly used techniques for image preprocessing mainly include gray-scale treatment, image tile correction, median filtering, etc. In this paper, three consecutive frames of an image are captured from the acquired video stream, while three consecutive frames of a license plate image are processed separately by gray level and vertical pixel gray level projection is performed. Second, the grayscale values of the two adjacent frames of the license plate image are subtracted, and then the vertical extent of the middle frame of the vehicle image is determined by the difference in grayscale between the two frames before and after the vehicle image. Finally, a horizontal gray-scale projection is made in the altered image, and the background image is subtracted to obtain the horizontal extent of the detected vehicle, which finally yields the accurate position of the moving target in complex conditions. And the final pre-processed picture for the next operation [10].

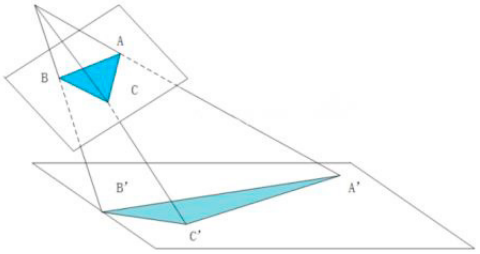


Fig.2. Image of the cropped license plate

4.License Plate Location

Traditional license plate location is to directly perform Sobel operation on the license plate, and then obtain the outline of the license plate region, or use a single color feature [11] or morphological feature [12] to intercept the outline of the license plate region. However, in complex environments, these localization methods suffer from large biases. In this paper, we adopt a scheme that combines image binarization based on color features, Sobel edge detection algorithm, and morphology algorithm to locate license plates.

4.1.Positioning Based on Color Features

The so-called color positioning is based on the RGB value of the license plate in the HSV(Hue, Saturation, value) color space corresponding to the color gamut value of the HSV color space cone diagram is shown in Fig.3 [7].

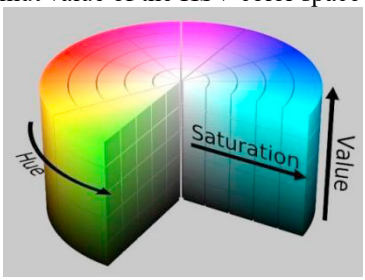


Fig.3. The HSV color space cone diagram

In order to find object information in image processing, RGB color models are generally not used, but RGB images are converted to HSV images, which is because RGB images do not well reflect object color information. Compared to RGB images, HSV color space can express color shading, hues and vividness very intuitively.

There are three parameters in HSV: H, S, and V, where H denotes Hue, S denotes Saturation, and V denotes Value. In OpenCV, the color parameter ranges are adjusted due to the limitations of the expressions: H ranges from 0 to 180,

S ranges from 0 to 255, and V ranges from 0 to 255. Since the images displayed by the imshow function in OpenCV are RGB color models by default, the desired effect can be achieved for each channel after separating the HSV channels.

4.2.Image Normalization

Gaussian filter is a linear filter that can suppress noise and smooth images in a limited way. It works similarly to the mean filter, which outputs the mean of the pixels in the filter window. The coefficient of the window template is different from that of the mean filter, which has the same template coefficient of one. The template coefficient of the Gaussian filter decreases with increasing distance from the template center. Therefore, the Gaussian filter is less fuzzy than the mean filter.

Before binarizing the image, we need to normalize the image using a Gaussian filter. Normalization is used to blur the image so that our objects can have an equal color distribution. Fig. 4 shows the image after the image normalization operation.



Fig.4. The image after image normalization operation

4.3.Image Binaryzation

Image binarization plays an important role in digital image processing, which can highlight the important outline and structure of an image after it has been converted to black and white, and greatly reducing the amount of algorithm calculation. Commonly used algorithms include the grand law algorithm and the iterative algorithm. In this paper, we propose to perform percolation operation in separated HSV channels for license plate recognition in complex environments.

The current license plate colors of our country are blue, green, yellow, etc. The blue and green channels in the image are extracted and the threshold for these two colors is set to be lower = np.array([40, 88, 40]) and upper = np.array([124, 255, 255]), respectively. The function cv2.inRange(hsv, lower, upper) is then used to apply the binary threshold to these two color channels within the given range of HSV values, converting the image to a binary image, and the resulting image is shown in Fig.5.



Fig.5. The original image and the image after HSV color conversion and binarization

4.4.Edge Detection

Edge detection is the key to license plate location. The purpose of edge detection is to identify points in a digital image with distinct brightness variations. Edge detection of an image will greatly reduce the amount of data, eliminate irrelevant information, and preserve important structural properties of the image. In license plate recognition, the license plate area often only accounts for a small part of the image, we need to focus on the license plate area, and so we need to carry out edge detection of the license plate image. Common edge detection algorithms include Roberts

operator, Prewitt operator, Sobel operator, Marr-Hidreth edge detection and canny operator. In this paper, Sobel operator and mathematical morphology are used to locate the license plate.

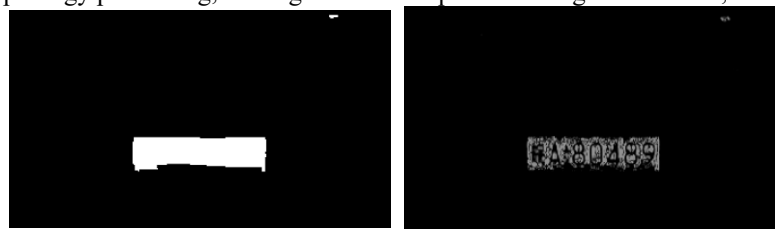
4.4.1.Sobel Algorithm

The Sobel operator is a popular treatment for edge detection. The Sobel operator weights the difference of the gray values in the upper, lower, left, and right fields of each pixel in the image, and reaches the extreme value at the edge to detect the edge. Technically, it is a discrete difference operator used to approximate the gradient of the image brightness function. Using this operator at any point in the image will produce the corresponding gradient vector or its normal vector. The Sobel operator not only yields better detection results, but also smooths the noise suppression, but the resulting edges are thicker and spurious edges may appear.

4.4.2.Mathematical Morphology Processing

Erosion and dilation operations are mainly used to fill in black-and-white blobs in binarized images, which makes the images sharper, smoother, and highlights the main objects. The first corrosion after expansion is an open operation, the first expansion after corrosion is a closed operation. Open operations generally smooth the outline of an image, weaken narrow sections, and remove elongated prominences, edge burrs, and isolated spots. The closed operation can also smooth the outline of the image, but different from the open operation, the closed operation generally fuses narrow gaps and slender bends to fill the cracks and holes in the image, which plays a role of connecting and filling, and the main structure of the image remains unchanged.

After the edge detection operation of Sobel operator is performed on the license plate recognition image, the threshold function is needed to convert the image from gray image to binary image, and then the morphology closure operation is performed on the image. The transformed image is shown in Fig.6 (a). After Sobel operator edge detection and mathematical morphology processing, the edge detection operation image is obtained, as shown in Fig.6 (b).



(a) Image after morphological closure operation **(b)** Image after edge detection operation

Fig.6. Image after edge detection processing

5.Character Segmentation

The vertical projection method is used to segment characters [8]. In this method, the number of black dots of pixels on the image perpendicular to the line (axis) is counted, and summed as the value of the position of the axis. The image cutting based on image projection is to map the image into this feature, determine the cutting position (coordinate) of the image based on this feature, and use this coordinate to cut the original image to get the target image. The segmentation effect is shown in Fig.7.



Fig.7. The effect of character segmentation

6.SVM Used in Recognize License Plate Character

Compared with the commonly used template-matching and neural network-based character recognition methods, the multi-class classifier based on support vector machine (SVM) can improve the accuracy of character recognition, save time, and has good adaptability with higher application value. SVM initially only classifies samples into positive or negative classes, which is a classical binary classifier. However, the real life usually needs to solve the multi-class classification problem, so the key of SVM research is to extend the two-class classification to the multi-class classification, and use SVM to deal with the multi-class classification problem. There are about four existing methods: one-to-one, one-to-many, support vector machine decision tree method and maximum distance between classes method. The "one-to-many" method is relatively simple and easy to implement, so this paper chooses the one-to-many method to design the classifier and recognize the photo character.

The one-to-many approach is to design a classifier by converting a multi-class classification into a two-class classification. The basic idea is to take the samples that need to be classified as one class, and combine the remaining samples as another class, and the problem will be transformed from a multi-class classification problem to several two-class classification problems. Suppose that the problem to be classified falls into three categories, "1", "2" and "3", and assume that they are separated by three rays originating from a point, as shown in Fig 8.

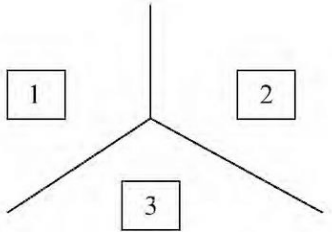


Fig.8. Sample classification chart

The following are the corresponding classification methods:

(1) Take "1" as a positive class, and "2" and "3" together as a negative class. The "1" is then divided using the decision function $f^1 = sgn(g1(x))$ constructed by SVM.

(2) Similarly, the decision numbers $f^2 = sgn(g2(x))$ and $f^3 = sgn2(g(x))$ are constructed respectively, and "2" and "3" are divided. If this formula can be accurately inversed to the actual situation, then it is easy to classify any input x, and one of the decision functions f^1, f^2 and f^3 is positive.

(3) If there is an error in the constructed decision function, the situation shown in Fig.9 may occur. In this case, according to the x belongs to $g1(x)$, $g2(x)$ and $g3(x)$ take the maximum value of the corresponding class.

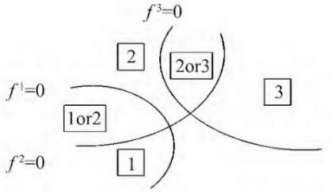


Fig.9. Error occurs in the decision function

This is generalized to the problem of multi-class classification, if there are *m* categories, the one-to-many method is to use SVM to construct *m* sub-classifiers. For the *h*-class subclassifier, those data in the sample that belong to the class are marked as positive, and the others are marked as negative, and then trained.

7.Verification and Analysis

In PyCharm platform, 390 sample license plate images taken under various complex conditions were selected for

character recognition of license plates. The correct recognition rate of the license plate was 96.52%. The recognition results are shown in Table 1.

Table 1. License plate recognition result

	Chinese character	English character	Number	Total character counter
Number of character	390	1271	1069	2730
The correct number of character	371	1232	1032	2635
Rate of identification	95.12%	96.93%	96.53%	96.52%

In addition, based on PyQt5, a simple GUI interface is designed for visualization, which can display the original image, the segmented image of the license plate after positioning, the image of the license plate after preprocessing, the result of license plate recognition and the total time required for recognition, as shown in Fig.10. Recognition time can be recorded while detecting whether the recognition is accurate or not, thus facilitating future algorithmic improvements.

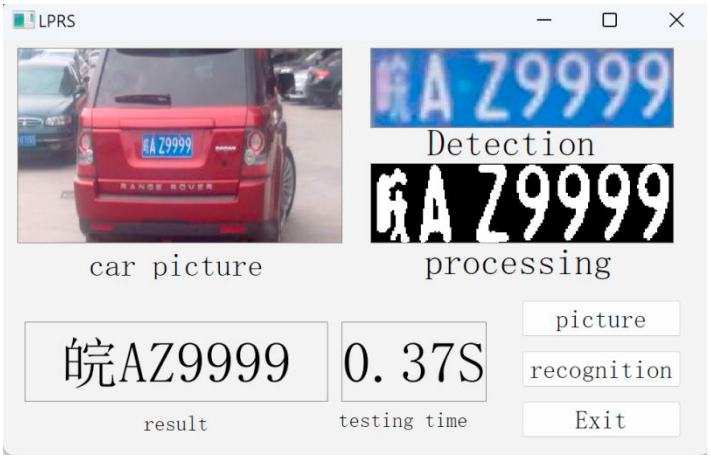


Fig.10. A simple GUI for user to operate

8.Conclusion

In summary, we have proposed a license plate recognition method based on OpenCV and SVM to solve the difficulties in the location and recognition of license plate targets caused by changing weather conditions and changing light intensity throughout the day in natural scenes. First, the Sobel edge detection algorithm combined with mathematical morphology processing was used to determine the approximate outline of the license plate. The fusion algorithm has a certain anti-interference ability to noise with fast calculation speed and can detect a relatively thin edge of the license plate. Then, the characters in the license plate were segmented by projection method. Finally, SVM classifier was used to recognize the segmented characters and output the recognition results. The proposed algorithm avoids the traditional process from induction to deduction, and realizes the efficient "transduction inference" from training samples to forecast samples, greatly simplifying the usual classification and regression problems. We used the PyCharm platform for verification, and selected 390 license plate sample pictures taken under various complex conditions, and the correct rate of license plate location is 96.52%. It is concluded that the proposed license plate recognition system can be applied in more complex environments, preserves the boundary features of the original

images well, effectively improves the recognition accuracy of low-quality images, and can be applied to license plate target location studies in rain and fog conditions.

Acknowledgment

The authors would like to thank Fuyang Normal University for its support in the experiment. This study is support -ed in part by the Anhui Provincial Natural Science Fundation (No.1608085QF159), the Key project of Natural Science Research in Universities of Anhui Province (No.KJ2020A0536 ， No.KJ2021A0657, No.KJ2023A0536), National College Students' innovation and entrepreneurship training program (No.202310371002).

References

[1] Sharma, N.,Dahiya, PK.,& Marwah, BR. (2022) "Performance comparison of various techniques for automatic licence plate recognition systems" ,*International Journal of Cloud Computing.***Volume 11 , Issue 2.2022**.PP.138-156.

[2] LIANGDong, GAOJun, CAOWei, FUQizhong, & ZHAOJing(2003)"The Location Algorithm of the Inclined License Plates Based on Mathematical Morphology and Orientation Field", *Chinese Journal of Electronics.***Volume 9,Issue 2.2003.**PP 163-166.

[3] Wang Zhongli, Ma Xiping, &Huang Wenlin(2019)"Vehicle License Plate Recognition Based on Wavelet Transform and Vertical Edge Matching", *International Journal of Pattern Recognition and Artificial Intelligence.* **Volume 34 , Issue 06 . 2019.** PP 2050016-2050016.

[4] Gao Huixin, Zhou Gang, Cao Yang, Luo Zhiyuan, Shen Zhicheng, & Gnana,MAJ.(2022)"Research on Edge Detection and Image Segmentation of Cabinet Region Based on Edge Computing Joint Image Detection Algorithm", *International Journal of Reliability, Quality and Safety Engineering.* **Volume 29 , Issue 05.2022.**

[5] Lili Han,Yimin Tian,& Qianhui Qi(2020)"Research on edge detection algorithm based on improved sobel operator", *MATEC Web of Conferences.* **Volume 309 , Issue 2. 2020.** PP 03031.

[6] Baiju,KB., Sabna,TS., & Lajish,VL.(2020) "Segmentation of Malayalam Handwritten Characters into Pattern Primitives and Recognition using SVM",*International Journal of Engineering and Advanced Technology.* **Volume 9 , Issue 3 . 2020.** PP 1817-1822.

[7] Pablo,FV., Daniel,G.,Javier,C.,& Javier,M.(2022)"New Aggregation Approaches with HSV to Color Edge Detection", *International Journal of Computational Intelligence Systems.***Volume 15 , Issue 1 . 2022.**

[8] ElMamoun ,M., Mahmoud,Z., & Kaddour,S.(2019)"Efficient Analysis of Vertical Projection Histogram to Segment Arabic Handwritten Characters", *Computers, Materials & Continua.* **Volume 60 , Issue 1.2019.** PP 55-66.

[9] Yu Jin,Gao Nan,Meng Zhaozong,& Zhang Zonghua(2021)"High-accuracy projector calibration method for fringe projection profilometry considering perspective transformation", *Optics express.* **Volume 29 , Issue 10 . 2021.** PP 15053-15066.

[10] Din,M., Bashir,A., Basit,A.,& Lakho,S.(2020) "Abandoned Object Detection using Frame Differencing and Background Subtraction", *International Journal of Advanced Computer Science and Applications* . **Volume 11 , Issue 7 . 2020.**

[11] Unnikrishnan,AP., Romeo,R.,& Rawther,FA.(2016)"License Plate Localization Using Genetic Algorithm including Color Feature Extraction", *Procedia Technology.* **Volume 24 , Issue C . 2016.** PP 1445-1451.

[12] Dogun Kim, Jin Kim, Eunil Park， AFA-Net: Adaptive Feature Attention Network in image deblurring and super-resolution for improving license plate recognition. Comput. Vis. Image Underst. 238: 103879 (2024)